

Requirement Analysis Report

E-Commerce Order Management Database System — Week 1 Deliverable

1. Introduction

This report presents the requirement analysis carried out for the proposed E-Commerce Order Management Database System. The purpose of this analysis is to understand the day-to-day business operations of an online shopping platform and to translate those operations into a structured set of requirements that will guide the database design, normalization, and implementation activities planned for the remainder of the semester project.

2. Business Scenario Analysis

The organization under study is a rapidly growing e-commerce company that currently manages its operations across multiple disconnected spreadsheets and manual records. As the customer base and product catalogue expand, this approach has become error-prone, difficult to scale, and unable to support timely business reporting. The company therefore requires a centralized relational database that can support the full lifecycle of an online order — from customer registration and product browsing through to payment, shipment, and post-purchase feedback.

Analysis of the scenario shows that the system revolves around three core activity clusters. The first cluster covers **customer-facing operations**: registration, product discovery, and product reviews. The second cluster covers **catalogue and inventory operations**: organizing products into categories and tracking stock levels supplied by vendors. The third cluster covers **transaction operations**: order placement, order line-item detail, payment collection, and shipment/delivery tracking. A well-designed database must keep these three clusters consistent with one another — for example, an order cannot be shipped before it is paid for, and stock quantities must be reduced only once a valid order is confirmed.

The scenario also implies a reporting need: management wants to derive meaningful business intelligence, such as best-selling products, revenue trends, delivery performance, and customer satisfaction, directly from the data captured during normal operations. This requirement influences the analysis by ensuring that every entity captures enough detail (dates, statuses, amounts) to support downstream aggregation and reporting without needing additional data entry.

3. Purpose of Core Entities

The system is built around nine mandatory core entities. The purpose of each entity in the overall business process is explained below.

Entity	Purpose in the System
Customer	Stores the personal and contact details of every registered user of the platform. It is the anchor entity for identifying who places an order and who writes a review.

Entity	Purpose in the System
Category	Groups products into logical classifications (e.g. Electronics, Apparel) so that customers can browse the catalogue efficiently and management can analyze sales by product group.
Product	Represents an item offered for sale, including its price and available stock. It links to a Category and is referenced whenever a product is ordered or reviewed.
Supplier	Records the vendors who supply products to the company, enabling procurement and restocking decisions to be traced back to a responsible party.
Order	Represents a single purchase transaction initiated by a customer. It captures the date of purchase, the overall amount, and the current status of the order (e.g. Pending, Shipped, Delivered, Cancelled).
Order Details	Acts as the line-item breakdown of an Order, recording which products were purchased, in what quantity, and at what unit price at the time of sale. This resolves the many-to-many relationship between Order and Product.
Payment	Captures how and when an Order was paid for, along with the payment status, so that finance operations and order fulfilment can be reconciled.
Shipment	Tracks the delivery of an Order to the customer's address, including the shipment date and delivery status, supporting logistics and customer service.
Review	Stores customer feedback and ratings for products they have purchased, supporting product quality tracking and helping future customers make informed decisions.

4. Major Business Processes

Based on the scenario and the core entities, the following major business processes have been identified as central to the operation of the platform.

Customer Registration & Profile Management

A new user supplies personal and contact details to create an account, and may update this information over time.

Product & Category Management

Administrators create and maintain product categories, and add, update, or discontinue products, including linking each product to a supplier and a category.

Inventory / Stock Management

Stock quantity for each product is monitored and adjusted as new stock is received from suppliers and as stock is consumed by customer orders.

Order Placement & Processing

A customer selects one or more products, an Order record is created, and the corresponding Order Details lines are generated with quantity and unit price for each product selected.

Payment Processing

Once an order is placed, a payment record is created capturing the payment method and status; the order can only progress once payment is confirmed.

Shipment & Delivery Tracking

After payment confirmation, a shipment record is created and its delivery status is updated until the order reaches the customer.

Customer Review & Feedback

After a product has been delivered, the customer may submit a rating and comments, which are stored against both the customer and the product.

Business Reporting & Analytics

Management periodically reviews aggregated data — sales volume, revenue, popular products, delivery performance, and customer satisfaction — drawn from the operational entities above.

5. Stakeholder Analysis

The table below identifies the stakeholders who interact with, depend on, or influence the E-Commerce Order Management Database System, along with their respective roles.

Stakeholder	Role / Responsibility
Customer	End-user of the platform who registers an account, browses products, places orders, makes payments, tracks shipments, and submits reviews.
Sales / Catalogue Administrator	Maintains product and category information, sets pricing, and ensures the catalogue reflects current offerings.
Inventory / Warehouse Manager	Monitors stock quantities, coordinates with suppliers for replenishment, and ensures products remain available for sale.
Supplier / Vendor	External party that supplies products to the company; their contact details are maintained to support procurement.
Order Fulfilment Staff	Processes placed orders, verifies payment status, and hands off confirmed orders to the shipment process.
Finance / Accounts Team	Reconciles payment records against orders and monitors revenue and outstanding payments.
Logistics / Shipment Partner	Manages the physical delivery of orders and updates shipment and delivery status.
Customer Support Representative	Uses order, payment, and shipment data to resolve customer queries and complaints.
Business / Management Team	Consumes reports generated from the database to make strategic decisions on pricing, promotions, and supplier relationships.
Database Administrator / Development Team	Designs, implements, secures, and maintains the database system itself, ensuring data integrity and performance.

Prepared as part of Week 1 — Understanding the Business Problem, E-Commerce Order Management Database System semester project.